

Are Weyl Gravity's Ghosts Real? Building Model Universes to Find Out

By GPT-5.6-sol and Asger Alstrup Palm

Einstein's theory says that gravity is the shape of spacetime. Matter bends spacetime; the resulting geometry guides planets, light and clocks. This picture explains gravitational redshift, the bending of light, gravitational waves, black holes and the expansion of the universe.

Weyl gravity begins with a more symmetrical set of equations. Unlike Einstein's equations, they contain four derivatives of the geometry rather than two. That feature makes the theory interesting as a possible route toward quantum gravity, but it also creates additional gravitational solutions. In ordinary treatments, some of those solutions carry the wrong sign. Physicists call them *ghosts*.

A ghost in an equation is not automatically a ghost in nature. It may be a redundant description, a wave forbidden by a global constraint, a solution excluded by a physical boundary condition, or a genuine instability. Our project is building a sequence of exactly controlled model universes to find out which answer applies.

The rule is simple:

A pattern in an equation is not yet a physical object. It must survive the constraints, propagate causally, affect a possible observation, and remain consistent when interactions and quantum physics are included.

This article explains what those tests mean, what the project has learned, and what remains unknown. It is a progress report about a research programme, not a claim that Weyl gravity already describes our universe.

The ghost test

An equation can contain more solutions than nature contains physical states. Before calling an extra solution a particle—or declaring it fatal—we ask six questions.

1. **Is it only gauge?** Different mathematical descriptions may represent the same physical situation.
2. **Does it satisfy the constraints?** A solution of the easiest linear equation may fail the full set of gravitational balance laws.
3. **Does it propagate causally?** A source in the future must not change a detector in the past.
4. **Can it affect an observation?** A physical direction should couple to a clock, detector, flux or another relational measurement.
5. **Does it survive interactions?** A small wave must fit into the nonlinear theory without violating a constraint or growing uncontrollably.
6. **Can it support a consistent quantum state?** Quantum probabilities must be positive after all redundant descriptions have been removed, and the quantum gauge identities must remain valid.

These are separate tests. Passing one does not silently grant the others. A classical wave is not yet a particle. A nonzero classical sign is not yet a quantum probability. A regular black-hole solution is not yet a stable black hole.

Gauge: when two descriptions are one reality

Gauge is the physicists' word for redundancy in a description. The same place on Earth can be described by latitude and longitude, a street map or a rotated grid. The numbers change, but the place does not.

Gravity has the same feature. Relabeling the coordinate grid changes the components used to describe spacetime without necessarily changing spacetime itself. Weyl gravity has an additional freedom: one may change the local scale—the choice of ruler—while preserving the light cones.

A gauge transformation is therefore not a new event, force or wave. Counting it as physical would count the same universe more than once. But a transformation that resembles gauge can instead be a physical symmetry. The difference is a measurable generator or charge, such as energy or angular momentum. The calculation must decide which case applies.

This matters especially for time. Saying that a shift of the *overall time label* is gauge in one closed, zero-total-charge model does not mean that clocks stop or change is unreal. Internal clocks can keep changing, and one can ask what another field does when a clock reads a particular value. In a larger sector where the time shift carries energy, the same transformation is a physical symmetry rather than gauge.

Causality: tomorrow must not alter yesterday

A causal theory does not allow a future event to change the past. If a source is switched on tomorrow, a detector must not respond today. A disturbance may influence only events inside its future light cone.

The physical response used for this test is the *retarded* response: it begins at the source and travels toward the future. Physicists also construct an *advanced* mathematical partner. It is useful for checking the equations and building comparison rules, but it does not authorize backward-in-time signalling. The physical source-to-detector rule remains retarded.

For gravity, it is not enough to test one convenient wave equation. The constraints and gauge relations must travel consistently with the wave. One of this project’s main technical achievements is to test causality at that full-system level.

Several model universes, not one unfinished cosmos

No single exactly solvable background answers every question. We therefore use several mathematical laboratories. Each removes some complications so that one issue can be tested exactly.

Model laboratory	What it is used to test
Closed spherical universe	Complete classical causal propagation, gauge reduction and candidate free quantum states. Space is finite but has no edge, like the surface of Earth one dimension higher. Its history is called the conformal cylinder.
Berger clock universe	Matter clocks, causal light signals, redshift and the first interaction calculations. Its spatial sphere is deliberately deformed in a controlled way.
Compact circle-by-sphere universe	Global balance conditions, nonlinear continuation and resonances of ordinary and additional gravitational waves coupled to electromagnetism.
Nariai curved universe	Whether the causal construction survives on a geometrically different curved background and under a first controlled deformation.
Schwarzschild exterior	Horizons, black-hole gravitational waves and whether familiar endpoint conditions remove the additional fourth-order family.

Results do not automatically transfer between these laboratories. A wave that is constrained in the compact circle-by-sphere universe has not thereby been removed from a Schwarzschild exterior. A boundary condition that works in a Euclidean calculation is not automatically a causal real-time rule. The one-loop quantum calculations use an additional Euclidean spherical laboratory: a calculation tool, not a real-time universe with observers. Every public claim is therefore labeled by its theory, background, charge sector, boundary conditions and stage of completion.

Three things we have learned

1. Causality belongs to the whole gauge system

The project has constructed exact classical retarded and advanced responses for the complete linked gravitational system on the closed spherical background. Separate constructions pass the same causal test in the Berger clock universe and in the Nariai curved universe. The constraints propagate with the fields: within those declared models, future sources do not alter past responses.

The work also extracts a reusable mathematical lesson. A complicated gauge theory can sometimes be embedded in a larger system whose causal behavior is easier to control. If the reduction back to the desired variables is local and preserves the constraints and physical comparison rule, the causal response can be transferred to the reduced theory. In plain language, one should judge causality using the complete network of fields, equations and redundancies—not necessarily one isolated fourth-order equation.

This is a classical result. It does not yet provide a complete interacting quantum theory, and it has not been proved for every curved spacetime.

The Berger laboratory adds something more tangible: a matter field acts as a clock, and an electromagnetic signal is emitted during a bounded time interval. Its response is retarded, and the frequency measured relative to the clock is unaffected by arbitrary coordinate, scale and electromagnetic gauge choices. One exact fixture gives a redshift ratio of $(1+z=2)$.

That is a real relational redshift calculation in a model, but not yet a prediction for a star or galaxy. The current observer programme is adding localized emitters, receivers, recoil and backreaction so that the global demonstration can become a more realistic measurement protocol.

2. Nonlinear balance removes some waves—and resonance adds another test

Linear equations describe very small disturbances. Nature is nonlinear: waves gravitate, light carries energy, and different fields interact. A linear solution is physically meaningful only if it can be continued into the nonlinear equations.

In the compact circle-by-sphere universe, the project has catalogued the certified finite families of ordinary Einstein-like waves, additional fourth-order waves and several exceptional global motions. At second order, five global balance conditions decide whether a finite combination has a formal continuation when controlled polynomial-in-time corrections are allowed.

This resembles balancing a spinning object. An individual motion may carry a nonzero global imbalance and fail, while a carefully chosen combination can cancel it. Some extra waves therefore disappear not because they were gauge, but because they cannot be the first step of an exact solution in that fixed charge sector.

Formal continuation is not the same as stable repeating motion. A correction may grow like time multiplied by an oscillation. The equations can still be solved, but the original small-wave approximation eventually breaks down. The project has found additional resonance conditions—forcing at a natural wave frequency—that are independent of the five global balances. Thus the emerging rule is:

Global balance decides whether a formal next step exists; resonance helps decide whether that next step can remain bounded.

The interaction story has also produced a useful correction. An apparent gravity–light interaction survived an initial reduction. Once the missing gauge structure was included, however, that term could be removed through the first derivative order. A larger second-derivative calculation is now testing whether any invariant interaction remains. This is precisely why the project treats intermediate symbolic results as provisional until the full gauge structure has been checked.

These findings do not yet prove long-term stability. They turn a vague fear about higher derivatives into explicit balance and resonance tests that can pass or fail mode by mode.

3. Black-hole regularity does not trivially erase the additional family

The black-hole programme begins with an exact family of spherical solutions of pure Weyl gravity. It includes Einstein-like members and genuinely additional fourth-order geometries. In the declared static setting, the project obtains regular horizons, a consistent energy, horizon entropy and an exact first-law relation.

The more difficult question concerns waves around the Schwarzschild member. The familiar Einstein axial gravitational wave is recovered, and the additional fourth-order content can be described through a curvature wave. That additional family has regular ingoing solutions at the future horizon.

The action-derived classical pairing gives an unusual result. The ordinary Einstein wave is self-silent in the pure-Weyl pairing, while controlled real-frequency calculations find nonzero pairing between the Einstein and additional waves and for a specified reconstruction of the additional metric wave. This shows that the additional solution is not merely a duplicated factor in the equation. The sign is a classical structural fact, not yet a positive quantum probability or energy statement.

The asymptotic conclusion is deliberately narrower than an earlier draft. The additional curvature wave has the same *leading* real-frequency characteristic as the Einstein wave in the tests completed so far. Horizon analyticity and that leading outer test do not force it to vanish. But a repeated characteristic can hide growing or logarithmic partners. The full metric falloff, finite flux at infinity and complete scattering problem have not yet been established.

The supported conclusion is therefore:

The additional axial family reaches the horizon regularly and is not removed by the endpoint tests performed so far. Whether a complete asymptotically flat boundary condition retains or excludes it remains open.

There is also a perfectly local linear initial-data restriction that sets the extra curvature and its initial velocity to zero. The unanswered questions are whether that restriction is physically selected, whether it gives the right nondegenerate wave space, and whether interactions preserve it. Polar waves, complex frequencies, ringdown, stability and Hawking radiation remain future work.

The quantum fork

Quantum physics may decide the ghost question, but the project is not yet at a final quantum verdict.

The strict pure-Weyl theory develops a nonzero local anomaly at one loop. An anomaly means that a symmetry of the classical equations fails after the quantum calculation is regulated. Adding a conformal compensator can repair the tested local quantum identity at that order—but adding a compensator changes the theory. The honest conclusion is not “the anomaly is solved,” but that the strict and compensated theories occupy different branches of the programme.

On the closed spherical background, a reduced calculation gives another clue. After gauge and constraint reduction, the candidate isolated conformal-graviton class disappears. Two collective curvature classes remain, but they have not been interpreted as particles.

A separate reduced free-wave construction produces candidate two-point functions with the required short-distance behavior. The Einstein-like family has a positive sign, while two additional families have negative signs. This is a concrete warning. It is not yet a particle theorem because the complete quantum gauge reduction, interacting observables and positive probability space have not been built.

Several outcomes remain possible. The negative directions might be removed by the final constraints, paired differently in a valid quantum construction, confined by nonlinear conditions—or survive as genuine ghosts and rule out the theory.

A compact public scorecard

The programme is deliberately incomplete. The table below separates results that exist in a declared model from the physical conclusions still being tested.

Demonstrated in a model	Partly demonstrated	Open
Complete classical causal propagation on several controlled backgrounds	A causal clock-defined redshift signal, with realistic localized apparatus still under construction	Electrons, mass generation and Standard Model matter
Gauge reduction and exact classical comparison rules	Additional compact and black-hole wave families, with their final physical status unresolved	A positive interacting quantum state and physical particles
Global nonlinear balance conditions and explicit resonance obstructions	Static black-hole thermodynamics and horizon-reaching additional waves	Full black-hole scattering, stability, ringdown and Hawking radiation
A one-loop anomaly calculation and a compensator-based local repair in a changed theory	Candidate reduced quantum two-point functions with both positive and negative signs	Gravitational lensing, realistic cosmology, dark matter and dark energy

Light, gravitational waves, redshift and black holes therefore occur in precisely scoped mathematical forms. Photons, gravitons and electrons do not yet occur as established particles. Gravitational lensing, galaxy dynamics and cosmology are later observational tests, not current achievements.

The live universe-building map and the searchable residual atlas carry the detailed status and update as new certificates land.

How AI-assisted research is being tested

This is also an experiment in scientific method. The programme is directed by Asger Alstrup Palm, Honorary Professor at DTU Compute, whose field is computer science rather than physics. AI agents perform much of the detailed symbolic derivation, adversarial review, literature mapping, verification and drafting. Palm chooses the overall direction and retains editorial responsibility for the public claims. His DTU affiliation identifies his academic role; it does not imply that DTU endorses the project or its conclusions.

The calculations are too large and too sensitive to signs and conventions to manage reliably as hand algebra alone. Some contain hundreds of linked field and constraint components and more than one hundred thousand symbolic interaction terms. Size is not evidence. The important point is that the terms are generated from declared equations and then subjected to independent checks.

The repository functions like a symbolic experimental laboratory. A material claim is expected to carry:

1. its theory, background, charge sector, boundary conditions and claim level;
2. a machine-readable certificate recording inputs, assumptions, hashes, outputs and unresolved fields;
3. an independent verifier that checks the decisive identity rather than trusting the program that generated it;
4. deliberately damaged inputs that must fail, showing that the verifier is actually testing something;
5. a record of convention errors, negative results and withdrawn interpretations rather than a cleaned-up success story;
6. a clean, reproducible route from the archived calculation to the public paper.

This discipline has already mattered. Cyclic sign conventions in the interaction calculation required repair. A black-hole causal claim was narrowed when a repeated asymptotic root was examined more carefully. An apparent invariant gravity–light interaction became removable after the missing gauge direction was restored. These are not embarrassments to hide; they are examples of the accountability system doing its job.

Code cannot prove assumptions that were never encoded. Global existence, physical boundary conditions, experimental relevance and quantum interpretation must each pass their own tests. The system is designed to stop a local algebraic success from being silently advertised as a result about particles or nature.

What would count as success—or failure?

The programme does not need Weyl gravity to win.

One possible result is **Einstein recovery**: the extra solutions are removed by physically justified constraints, boundaries or quantum reduction, while ordinary relativity survives with the correct observables.

Another is **healthy new physics**: an additional wave, interaction or cosmological effect survives causality, nonlinear stability, observation and quantization. Only then would it become a credible prediction.

A third is **sector-dependent viability**: the theory works on a closed, zero-charge universe but not when time translation carries energy at an outer boundary, or it works only with an explicitly stated compensator or boundary condition. That is not a contradiction. It tells us which theory and which physical sector are actually under discussion.

The fourth outcome is an **exact no-go**: an unavoidable anomaly, negative physical state, instability or boundary obstruction ends the candidate theory under a complete set of assumptions.

The project can stop at any rung and still produce useful physics. A precise failure tells us which attractive idea cannot describe nature, under which conditions, and what a successful replacement would have to change. A passed rung narrows the alternatives and makes the next test meaningful.

The central question remains open:

Are Weyl gravity’s additional solutions redundant descriptions, constrained waves, boundary excitations, healthy new physics—or genuine ghosts?

We are building model universes carefully enough that each of those answers can be tested rather than assumed.

For readers who want the technical continuation

This article is the public front door. The physicist executive summary provides the claim-by-claim result spine, exact scope labels and links to papers and certificates. A PDF version is available as a short technical briefing.

The public construction map shows dependencies between familiar physical milestones. The residual atlas provides the detailed, searchable ledger of wave families and their causal, symplectic, nonlinear, observational and quantum status.